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Anisotropy in the Hubble parameter and large-scale cosmological inhomogeneity*

A. J. Fennelly *Physics and Astronomy Department, Western Kentucky University, Bowling Green, Kentucky 42101, USA*

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Summary. The reported anisotropy of the Hubble expansion parameter is analysed within the context of a possible large-scale inhomogeneity. We are able to fit the data to a tilted self-similar cosmology with a Gaussian density distribution superposed on a homogeneous Universe. The particular model sets the inhomogeneous region's diameter at 540 Mpc, with our Galaxy 120 Mpc from the centre which is toward $\alpha = 12^h 42^m$, $\delta = +20°$. This eliminates the conflict between the anomaly and the isotropy of the microwave background.

1 Introduction

The expansion of the Universe has been generally considered to be homogeneous and isotropic to high degree [1], as indicated, for example, by observations of galaxies [2] and analysis of the microwave background radiation [3]. However, evidence on Sc I galaxies has suggested that this may not be so on large scales (~120 Mpc) [4], and some confirmation of this result has come from data on compact galaxies with absorption spectra and on Markarian galaxies [5], the brightest cluster galaxies [5, 6], Type I supernovae [7], the van den Bergh galaxy luminosity classification [8], and radio galaxies associated with galaxy clusters [9].

More recently, a thorough analysis has established the existence of a Hubble parameter anomaly more firmly, both for a variation with direction and an effect due to intervening galaxy clusters [10]. Southern hemisphere photometry [11] has completed the original Sc I map [4]. A complete sample of Sc I–Sc II galaxies has provided a map of the anomaly both for optical velocities and radio velocities [12]. Subsequent interpretations do not agree [10, 13], but application of one has substantially reduced the scatter in a published Hubble diagram for nuclear magnitudes of first-ranked cluster galaxies [14].

An interpretation of the anomaly as due to a peculiar velocity of 500 ± 125 km/s of the Galaxy presents a clear discrepancy with the isotropy of the microwave background radiation [3], which limits our peculiar velocity relative to the frame of last scattering of the radiation to 250 km/s. This is a most serious disagreement within the context of what is generally believed about the Universe.

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Several interpretations and explanations have been advanced, ranging from the exotic to the ordinary: One group of observers has developed a new physical theory requiring a universal low-mass scalar boson field interacting with finite-mass photons [15]. This leads to a static hierarchical Universe with 'tired-light' redshifts, in which photons traversing a cluster of galaxies experience a greater redshift due to photon-boson scattering. It can explain one interpretation of the data [10] but is close to an 'ether' theory of light propagation, and will have to be tested experimentally before it can be taken seriously.

Other explanations involve selection effects or systematic deviations in populations or intrinsic properties of objects across the celestial sphere. These seem not to be the case [13], and there remain only dynamic explanations: a systematic peculiar velocity of our Galaxy, a systematic anisotropy in the Hubble expansion (both of which can exist in homogeneous Universes), or alternatively an inhomogeneity.

Inhomogeneity of the purely Swiss-cheese type, affecting galaxies behind clusters of galaxies can be ruled out [16]. Doroshkevich & Shandarin [17] have presented a model of the formation of galaxies and clusters of galaxies that generates an anisotropy in H consistent with the data. It is purely local due to the velocity perturbations induced during galaxy formation and covers scales of 50–100 Mpc, with velocities 500–1000 km/s. A peculiar velocity of the galaxy of 600 ± 125 km/s toward $\alpha = 32^\circ \pm 20^\circ$, $\delta = +53^\circ \pm 11^\circ$ is consistent with the data [13]. A systematic angular gradient in H was rejected in [13] by applying Occam's razor and accepting the simplest explanation. There seems to be no advantage in that anyway, since the conflict with the microwave background radiation would remain: we would have exchanged a discrepant Doppler shift for a discrepant shear in a homogeneous Universe.

But what of an inhomogeneous Universe? We examine here an alternative that is inhomogeneous and produces an angular gradient of H on the celestial sphere. The degree of inhomogeneity is within the experimental errors of [13] for their near sample of galaxies (at $\pm 3\sigma$), which they used as evidence for a peculiar velocity. We are guided in this by the results of Guthrie [6], which show the effect decreasing closer to us than ~100 Mpc and further from us than ~140 Mpc.

Mavrides [18] has modelled a general class of inhomogeneous cosmologies which provide order-of-magnitude arguments in support of this. He imbeds an Einstein–de Sitter universe within a spherical vacuum region which is joined to an exterior Einstein–de Sitter model using the Swiss-cheese junction conditions. The two Einstein–de Sitter regions have different expansion rates and so their fitting must use an intermediate vacuum shell. A Tolman–Bondi solution is given, analytically continued through the three regions. While this is a desirable first step, it seems that a more specific model could be fitted. First there is Guthrie's result [6] which implies rather more structure than two homogeneous models fitted together through a vacuole. Then an observation of Dodd *et al.* [19] shows that there may be a change in the number count–diameter relation in the opposite direction on the celestial sphere, which may indicate a limit to this inhomogeneous structure at a particular radius; in such a case there would be no intermediate vacuole, but rather a clump in a Friedman background.

The Dodd result may not be statistically significant [20] but it does suggest a model with a definite structure to be fitted with the data. First, the detailed nature of the reported anisotropy will be examined and this will lead to a map of a possible inhomogeneity in which we may exist. The structure of such an inhomogeneity will then be studied in a few dynamical models. Finally its effects on the microwave background radiation will be determined. Throughout the discussion we shall use the following terms in reference to distance r: local for $r \sim 10$ Mpc, large-scale for $r \sim 100$ Mpc and global for $r \sim 1000$ Mpc.

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2 The anisotropy model

According to the reported observations the anisotropy is maximized at a distance modulus for Sc I spiral galaxies of 35.7 which, for an absolute magnitude of −21.2, corresponds to a luminosity distance of 132 Mpc. For this distance the anisotropy is mapped in Fig. 1. Coordinates of right ascension and declination are marked, as is the galactic plane GP. The line DL separates regions of maximum redshift MAX from minimum redshift MIN. The hatched lines plot the anisotropy gradient in contours. The anisotropy gradient is less pronounced for greater and lesser luminosity distances, vanishing eventually.

Successive versions of Fig. 1 for a range of luminosity distances projected into a plane marked by the dotted line in Fig. 1 are shown in Fig. 2. The dotted circles are celestial spheres concentric about our Galaxy G. The dotted circle passing through O is that for

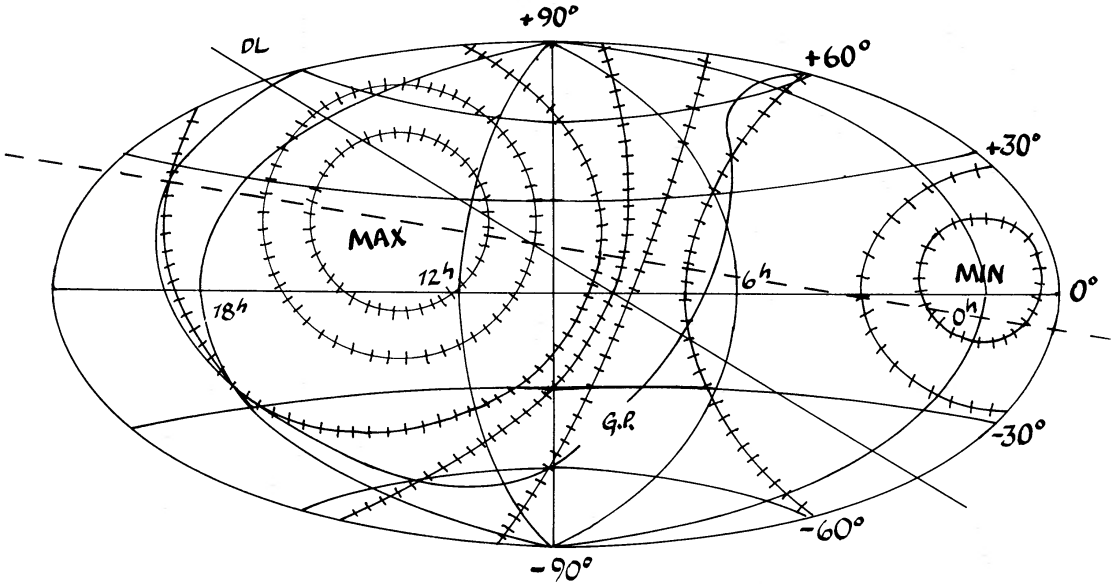


Figure 1. Representation of redshift gradient on celestial sphere. Regions of maximum and minimum redshift are marked MAX and MIN. Coordinates are right ascension and declination (1950.0). Galactic plane is G.P., line dividing regions is DL. Isocontours of redshift variation are ++++++ lines. Plane slide for Fig. 2 is marked – – – – –.

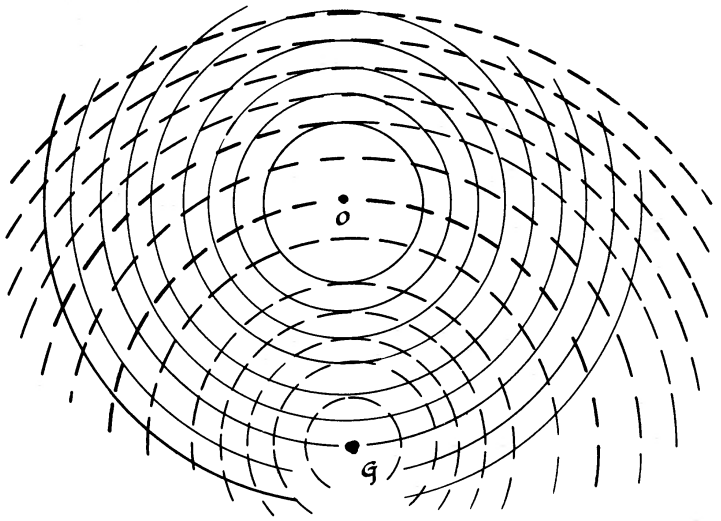


Figure 2. Plane from celestial sphere showing variation of redshift gradient with luminosity distance. Galaxy is at G, 'centre' of redshift isodents ——— is 0, luminosity isodents are – – – – –.

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$r \sim 132$ Mpc. The concentric circles about O intersect each celestial sphere at contours of anisotropy gradient and represent contours of increasing redshift gradient radial about O. The redshift of each galaxy may thus be considered as a combination of two factors, one due to expansion (mapped by the dotted circle) and the other due to an inhomogeneous structure about O (mapped by the circle about O). The sum of the two redshift components is given in Fig. 3.

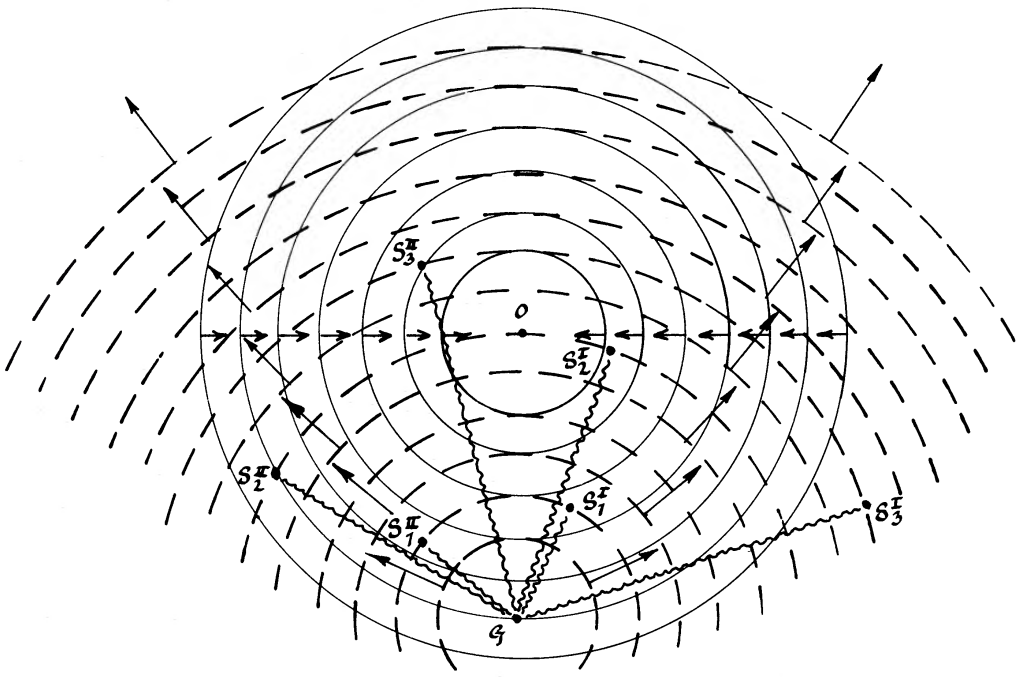


Figure 3. Redshift map for model suggested by Fig. 2. Small arrows → mark 'gravitational' energy shift direction, across isodents of δHM ———. Large arrows ⟶ mark expansion Doppler shift direction, across isodents of luminosity density. Source markings S_1, S_2, S_3 are placed at equal luminosity distance (subscript); source markings S^{I}, S^{II} are placed at different δHM isodents (superscripts). Photons are wavy lines from S_1^{I}, S_1^{II}, S_2^{I}, S_3^{I}, S_3^{II} to G. O is the centre of the system.

3 Dynamics

We shall first examine a spherically symmetric model. In Fig. 3 the paths of photons from sources S to our Galaxy G are represented by wavy lines. The dotted circles around G are contours of increasing Hubble expansion velocity, while the full circles about O are contours of decreasing gravitational shift. The arrows represent the direction of redshift. In $S_{\mathrm{a}}^{\mathrm{A}}$ the subscript refers to radius about G while the superscript refers to that about O. Thus S_2^{I} and S_2^{II} have equal Hubble expansion velocities but different gravitational velocity shifts, etc. Since there is essentially no difference between Doppler shifts and gravitational shifts [18] we could model the effects of the inhomogeneity as due to either one, the other entering indirectly through their coupling in the dynamics.

Since the centre of the homogeneity is ~120 Mpc away (toward $\alpha \sim 12^{\mathrm{h}}$, $\delta \sim 20°$), a diameter of the patch (embedded in the homogeneous expanding Universe) is $\gtrsim 300$ Mpc. Dodd *et al.* have reported evidence of inhomogeneity in the direction $\alpha \sim 2$–3^{h}, $\delta \sim -30°$ with a boundary ~150 Mpc [19]. While this may not be statistically significant [20] it may be real, giving a radius ~270 Mpc. This is about 10 per cent of the present horizon distance of a Friedmann universe, so relativistic dynamics must be used.

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We used a Tolman–Bondi [21] model as modified by Bonnor [22]. The redshift reduces to the formula (for inhomogeneous models)

$$(1+z)\,|_{S}^{G} = 1 + \Delta V/c + \phi_G - \phi_S, \tag{1}$$

where ΔV is the relative expansion velocity between G and S and $\phi_i = -GM(t)/c^2 r_i$ is the gravitational potential at r_i on a spherical surface enclosing the mass M(i). Assuming a linear expansion rate θ, and the relation $d = r_L/(1+z)^2$ connecting true distance d on spatial hypersurfaces labelled by constant time $t - t_0$ with luminosity distance r_L along the light cone (removing the effects of the redshift on our distance measurements) to find the distance of the emitter ($\Delta V = \theta d$), we may obtain information about ϕ_i if ϕ(G) can be determined. But ϕ_i may contain a contribution due to a peculiar velocity relative to a Friedman model (or to the centre of symmetry in a spherically symmetric model) that is yet of interest.

The existence of a peculiar velocity relative to a Friedman model must be expected even for a dust flow. The equation of motion is $a_\mu = u_{\mu\|a}u^a = 0$, which just says that the velocity decreases with time as $u = u_0 e^{-\alpha}$ in a Friedman universe where e^α is the expansion scale factor. But in an inhomogeneous universe it will have a different form which can be easily seen if we consider inhomogeneous self-similar universes such as those of Cahill & Taub [23] or Lin *et al.* [24]. For a self-similar spacetime with a tilted time coordinate we have a metric

$$ds^2 = e^{2\psi}\{-(dt - u_\mu\omega^\mu)^2 + (\delta_{\mu\nu}\omega^\mu\omega^\nu)\} \tag{2}$$

where ψ contains the information about the inhomogeneity (viz. $d\psi = b_\mu\sigma^\mu$ with b_μ a constant vector and σ^μ a basis of differential forms on the spatial hypersurfaces) and $\omega^\mu = e^\alpha e^\beta_{\mu\nu}\sigma^\nu$ (α and $\beta_{\mu\nu}$ functions of time describing the volume and shape of the matter ellipsoids) [25]. The conservation equation reads

$$\frac{\partial u_\mu}{\partial t} + u_\lambda(\theta_{\lambda\mu} + \tau_{\lambda\mu}) + e^{-\alpha-\psi}\, e^{-\beta}_{\mu\lambda} b_\lambda = 0, \tag{3}$$

where

$$\theta_{\lambda\mu} = \dot{\alpha}\delta_{\lambda\mu} + (e^\beta)'\, e^{-\beta}_{(\lambda\mu)}$$

and

$$\tau_{\lambda\mu} = (e^\beta)'\, e^{-\beta}_{[\lambda\mu]}$$

are the expansion tensor and rotation tensor of an observer's reference triad with respect to a set of Fermi-transported axes. The solution is (no sum on σ^μ_μ)

$$u_\mu = \exp\left[-\alpha - \int(\sigma^\mu_\mu + \sigma_{\mu\lambda})\,dt\right](u^0_\mu \cos\tau_{\mu\lambda}\,dt + u^0_\lambda \sin\tau_{\mu\lambda}\,dt)$$

$$-\exp\left[-\int(\theta+\tau)\,dt\right]_{\mu\lambda} \times \int \exp\left[-\alpha + \int(\theta+\tau)\,dt\right]_{\mu\lambda} e^{-\beta}_{\lambda\nu} b_\nu\,dt. \tag{4}$$

Clearly the inhomogeneity substantially changes the velocity in the second term on the right of equation (4).

The velocity will also cause an important apparent variation in physical quantities. For the redshift z a spatial gradient will appear. Using standard tensor transformation laws with

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the metric in equation (1) we find

$$\frac{\partial z}{\partial X^{\mu}} = \frac{\partial z}{\partial t} u_{\mu}, \tag{5}$$

an extra component to any spatial variation of z. Then

$$(1+z)\Big|_{S}^{G} = 1 + \frac{\Delta V}{C} + \phi_G - \phi_S + \int_S^G \frac{\partial z}{\partial t} u_{\mu}\, dx^{\mu}. \tag{6}$$

With metric

$$ds^2 = -(\omega^0)^2 + R'^2\, dr^2 + R^2(d\theta^2 + \sin^2\theta\, d\psi^2), \tag{7}$$

$R = R(r, t)$, Einstein's equations for dust with density ρ without cosmological constant, are ($' = \partial/\partial r$):

$$2RR'^2 = F(r) \quad \text{and} \quad 8\pi\rho = F'/2R'R^2, \tag{8}$$

with $F(r)$ a function of r only, and the continuity equation reads

$$\dot{\rho} = -\theta\rho = -\left(\frac{\dot{R}'}{R'} + \frac{2R'}{R}\right)\rho \tag{9}$$

and we specialize to the shearfree case for which the expansion θ depends only on time [26]. We can then let R be separable $R = T(t)\,P(r)$ and we choose a Gaussian density distribution function for the density ρ of amplitude δ and dispersion σ:

$$\rho = \rho(t)[1 + \delta e^{-r^2/\sigma^2}]. \tag{10}$$

Then equation (8) can be solved to give

$$R = (K/2)^{1/3}[3/(8\pi K)]\, t^{2/3}\, [1 + \delta e^{-r^2/\sigma^2}]^{-1}. \tag{11}$$

The mass within a radius r is

$$M = 4\pi \int_0^r \rho R' R^2\, dr, \tag{12}$$

giving

$$\Delta M(r) = \frac{2\pi\rho_0(2\delta + \delta^2)(1 - e^{-r^2/\sigma^2})}{(1+\delta)^2(1 + \delta e^{-r^2/\sigma^2})} \tag{13}$$

for the mass beyond that due to a homogeneous distribution.

The acceleration equation solved in the frame with $\omega^0 = dt - u_R dR = dt - u_r dr$ gives

$$u_r = u_r^0 (K/2)^{1/3}(3t/8\pi K)^{2/3}\, [1 + \delta e^{-r^2/\sigma^2}](-2\delta/\sigma^2)\, r. \tag{14}$$

Using the structure of the redshift variation we find $K = 4/9$, $u_r^0 \approx 200$ km/s, $\sigma \approx 36$ Mpc and $\delta \approx 0.03$.

A more exact model may be generated following the method of [23] and [24], which show that several forms of density variation are possible. The model may be extended to spheroidal self-similar systems using the technique of [25] and allowing more general shearing motions and space curvatures.

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One exotic model would involve the extreme case of very large δ in the density so that an extreme black hole would cause the necessary gravitational shift. The required mass would be $M \sim 10^{50}$ g to produce gravitational redshifts. A black hole of such a mass would be an X-ray source from the accretion of intergalactic matter. The X-ray luminosity L_x is given in terms of the mass accretion rate $\dot{M}_{acc}$ and the conversion efficiency ϵ as [27]

$$L_x = \left(\frac{\epsilon}{0.1}\right)\left(\frac{\dot{M}_{acc}}{M}\right) 10^{46}\,\text{erg/s} \tag{15}$$

and the Bondi [28] accretion rate is

$$\dot{M}_{acc} = \frac{8\pi(GM)^2}{(kT)^{3/2}}\,\mu^{3/2}\rho, \tag{16}$$

with k, μ, T, ρ, the Boltzmann constant and the gas mean molecular weight, temperature, and density, and M the black-hole mass. Assuming ionized hydrogen with $\rho \sim 10^{-31}\,\text{g cm}^{-3}$ and $T \sim 10^6$ K, one finds $\dot{M} \approx 1.1 \times 10^{29}$ g/s and $L_x \approx \epsilon(5.5) \times 10^{50}$ erg/s. For an efficiency of 10^{-3} that becomes 5.5×10^{47} erg/s and, at a distance ~ 120 Mpc, the received intensity of this source would be $I \sim 4 \times 10^{-7}\,\text{erg/(cm}^2\text{s)}$. The identification of such a source in the direction of maximum Hubble anisotropy would indeed be interesting.

4 The microwave background

The conflict of the Hubble anisotropy with microwave isotropy may now be considered. For this large-scale inhomogeneity the results of Dyer are applicable [29]. For a central mass of $\sim 10^{19} M_\odot$, there is a large redshift in the pattern centred on the inhomogeneity with $\delta T \sim -0.3$, and a hot ring (blueshift) of radius $\sim 30^\circ$ with $\delta T = 0.1$ per cent around it. If there is a central black hole of mass $\sim 10^{17} M_\odot$ then the cool point at $\alpha \sim 13^h$, $\delta \sim +20^\circ$ has $\delta T \sim -0.06$ per cent and a hot ring of radius 2 arcmin with $\delta T \sim +0.03$ per cent. Other perturbation schemes produce variations of the same order. We see that such an inhomogeneous structure may just barely be reconciled with the isotropy of the microwave background radiation.

This should not be too surprising. The effects of a large-scale inhomogeneity appear in the behaviour of geodesic congruences. The microwave photons propagate over global distances, spending a relatively small part of their lifetime in the region of the large-scale inhomogeneity. Also their momenta obey $p_\mu p^\mu = 0$, indicating a much weaker effect on their geodesics.

However, the galaxies spend their entire histories under the influence of local and large-scale dynamics with their momenta obeying $p_\mu p^\mu = -1$. Their geodesics are therefore far more seriously affected by the inhomogeneous region's dynamics. Also, not only are their images affected but their own motions as well, while only the image of the last scattering surface of the microwave background is involved. A Doppler velocity of 500 km/s [13] would appear in both of them and apparently it does not. The inhomogeneous model therefore explains the Hubble anisotropy and the background radiation isotropy. It is important to remember, however, that the microwave isotropy is not yet a whole-sky survey. Therefore, anisotropies may yet remain to be discovered which would have to be confronted by these models. It may well be that a significant anisotropy will require the addition of a rotational velocity u_ψ component, or of some form of electromagnetic fields. On the other hand such a discovery might also invalidate the models.

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Debney [30] has presented an interesting inhomogeneous cosmology. He shows that observers inside a black hole would largely see an anisotropic pattern of redshifts from sources distributed isotropically about them. The Hubble anisotropy [10, 13] may be possibly represented by this model, and it should be investigated. It is not yet clear how background radiation would appear, so the question of suitability remains open.

5 Other astrophysical effects

One expects other astrophysical effects to appear if we have such an eccentric position in a spherically symmetric Universe. Among these are the isotropy of number counts of galaxies, the apparent proper motions of objects, and the distribution of ellipticities of objects.

The shear of a bundle of null rays obeys an equation different from that obeyed by a time-like ray bundle's shear. This is

$$\dot{\sigma}_{\mu\nu} = -C_{\mu_0\nu_0} - \tfrac{2}{3}\theta\sigma_{\mu\nu} \tag{17}$$

for the rate of change of $\sigma_{\mu\nu}$ along the ray path. In the geometry considered above the solutions are

$$\sigma_{\mu\mu} = \sigma_0 \left.\frac{2}{R^2R'}\right|_{R_0}^{R} + \sigma_0 \left.\left(\frac{M}{R^2}\right)\right|_{R_0}^{R} \tag{18.1}$$

$$\sigma_{\theta\theta} = \sigma_{\psi\psi} = -\left.\frac{\sigma_0}{R^2R'}\right|_{R_0}^{R} - \sigma_0 \left.\left(\frac{M}{2R^2}\right)\right|_{R_0}^{R} \tag{18.2}$$

evaluated between the integration limits R_0 and R. The evolution of the ellipticity e along the ray path is given in terms of the rms shear $\sigma^2 = \tfrac{1}{2}\sigma_{\mu\nu}\sigma^{\mu\nu}$ as [31]

$$(\ln e)^{\cdot} = 2\sigma. \tag{19}$$

The apparent proper motion of an object with Cartesian position vector e_α is (20)

$$\dot{e}_\alpha = \sigma^\alpha_\beta e_\alpha - \sigma_{\lambda\delta}e^\lambda e^\delta e_\alpha. \tag{20}$$

For a reasonable inhomogeneity, both of these effects would be difficult to detect. Simple order-of-magnitude calculations give that $\sigma \sim 10^{-5}$/Mpc for close passage just outside the region of maximum density (~ 10 Mpc). Then, for a path traversing the regions of maximum Hubble anisotropy ($\sim 10^2$ Mpc), equation (19) gives an ellipticity change $\Delta \ln e \approx 2 \times 10^{-5}\,\Delta l = 2 \times 10^{-3}$, implying $\Delta e \approx 0.002$. Similarly equation (20) gives

$$\Delta \ln e_\alpha \approx (\sigma - \Sigma\sigma_m Y_{2m})\,\Delta l,$$

where $\Sigma\sigma_m Y_{2m}$ is the sum $\sigma_{\lambda\delta}e^\lambda e^\delta$ considered as a sum over harmonics on the unit sphere (20); then $\Delta e_\alpha \lesssim 0.002$ for the same case. But these same considerations may rule out the extreme black-hole case, for then $\sigma \sim -10^{-3}$/Mpc for an impact parameter ~ 1 Mpc. With a ray path $\Delta l \sim 10^2$ Mpc, $\Delta \ln e \sim 0.2$ implies $\Delta e \sim 0.22$ and $\Delta e_\alpha \lesssim 0.22$. These effects should be detected if they exist. It is doubtful that they do.

In the case of a non-exotic kind of inhomogeneity, rather subtle trends may be revealed in a data sample of sufficient statistical significance. Sample luminosity distances depend on the expansion, $\theta = \nabla^a K_a$, of a ray bundle which itself obeys the equation

$$\dot{\theta} = -R_{ab}K^aK^b - 2\sigma^2 - \theta^2/2 \tag{21}$$

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along the ray path. In equation (21) the part $-\theta^2/2 - R_{ab}K^aK^b$ is expected in any case, and the contribution of inhomogeneity to θ^2 is small. The change in $\dot{\theta}$ due to the inhomogeneity is $-2\sigma^2 - \Delta(R_{ab}K^aK^b)$ with $\Delta R_{ab} = 8(T_{ab} - \frac{1}{2}g_{ab}T) \sim \rho_0^{(t)}/R$ by equations (10) and (11). Then we write $\Delta\dot{\theta} \approx -2\sigma^2 - \rho_0/R$ and make dimensional estimates again, $\sigma^2 \sim 10^{-10}\,\mathrm{Mpc}^{-2}$ and $\Delta R_{ab}K^aK^b \sim 10^{-5}\,\mathrm{Mpc}^{-2}$. The change in θ over a 10^2 Mpc path is then $\sim 10^{-3}$/Mpc which could lead to a sizeable change in apparent luminosity over a sample of objects. An anisotropy in the luminosity function of galaxies should exist between separate hemispheres.

With regard to published studies, we note in passing that such an anisotropy may well exist for the luminosity function of galaxies [32] and a variation of ellipticities of galaxies may have been detected [33]. A systematic orientation of galaxies toward the cluster centre may exist in the Coma Cluster [34]; it is there that such an extreme black hole may be. These observations await confirmation and may well be due to a variety of other astrophysical effects.

We now turn to number counts. A suitable formula from Kristian & Sachs [20], to first order in gradients and direction cosines e_μ, is

$$dN = (1+z)\,r^2\,dr\,d\Omega\{n + r(n\theta + n_\mu e^\mu)\}, \tag{22}$$

with the redshift z given by

$$z = \left(\frac{\theta}{3} + \Sigma\sigma_m Y_{2m}\right) r + \left[\frac{\rho}{6} + \frac{\theta^2}{3} + e^\mu\left(\frac{3}{5}\theta_\mu + \frac{1}{5}\nabla^\lambda\sigma_{\mu\lambda}\right)\right]\frac{r^2}{2}. \tag{23}$$

Retaining only terms to first order in the luminosity distance r we have

$$dN = r^2\,dr\,d\Omega\,\{(n\theta + n_\mu e^\mu + n[\theta/3 + \Sigma\sigma_m Y_{2m}])\,r\}. \tag{24}$$

For the metric of equation (2) we have $n_\mu = \partial_\mu n + u_\mu\,\partial_0 n$ for a gradient, which causes an anisotropy in the source distribution, along with the $\Sigma\sigma_m Y_{2m}$ angular harmonic term. We integrate separately over both halves of the celestial sphere N_1, centred on $\alpha = 12^h42^m$, $\delta = +20°$, and N_2 centred at the antipole, again looking for dimensional estimates, we form the quantity $|N_1 - N_2|/(N_1 + N_2)$, finding

$$|N_1 - N_2|/(N_1 + N_2) \sim (\tfrac{3}{8}u\theta + \sigma + \nabla\ln n)\,r, \tag{25}$$

where the region surveyed extends to r and we evaluate $\nabla \ln n$ between the two halves of the celestial sphere. We have $3u\theta/8 \sim 10^{-4}$/Mpc and $\sigma \sim 10^{-5}$/Mpc. Next, from equation (10) we obtain $\nabla \ln n \sim 4.6 \times 10^{-5}\,\mathrm{Mpc}^{-2}\,r$. So we find overall that

$$|N_1 - N_2|/(N_1 + N_2) \sim 1.1 \times 10^{-4}\,\mathrm{Mpc}^{-1}\,r + 4.0 \times 10^{-5}\,\mathrm{Mpc}^{-2}\,r^2.$$

For a survey over $\sim 10^2$ Mpc we find the density difference is ~ 0.4601. A locally steep density gradient in the same direction does exist, but not on scales > 20 Mpc [35]. However, de Vaucouleurs [36] noted just such a large-scale gradient with the ratio $N_1/N_2 \sim 3.2$–4.1. The ratio implied above is $N_1/N_2 \sim 2.7$. This is a highly suggestive result. Other ways that this might manifest itself is in a change in the luminosity function of galaxies with a greater number of brighter galaxies in the N_1 region (about $\alpha = 12^h42^m$, $\delta = +20°$) as has been reported [32]. Or one might expect anomalous behaviour in the number count–angular diameter relation *a la* Dodd *et al.* [19].

We have not resolved the issue here but further data, particularly number counts, should decide if the model is to be rejected. As long as the supercluster/no supercluster discussion continues it is likely that this will also. One recalls that Omer's inhomogeneous model [37] still makes interesting reading.

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